

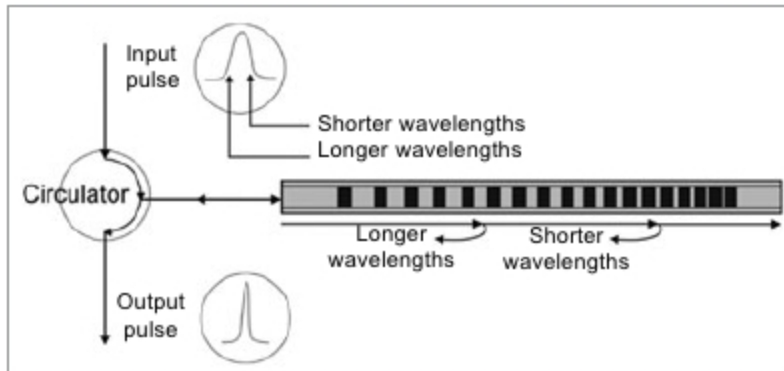


Fig. 3: Linearly-chirped grating

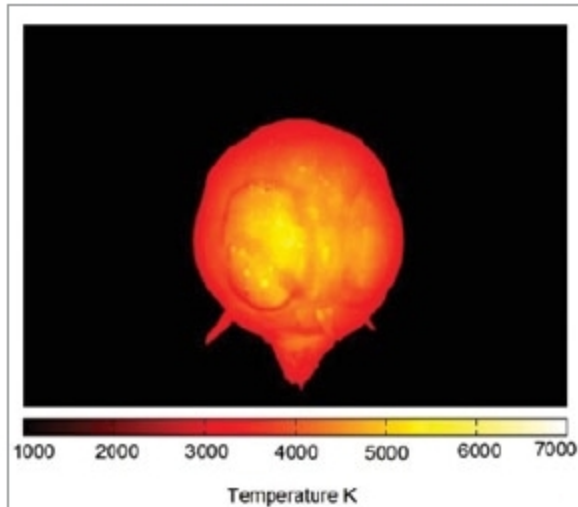


Fig. 4: Atomic emission spectrometry for fireball temperature measurement

tive index and length of grating. Fig. 2 shows the schematic of one such grating in which grating period is increasing linearly with the length of grating. In this figure, $\lambda_1 < \lambda_2 < \lambda_3 < \lambda_4$

Another linearly-chirped grating could have a linearly decreasing grating period (Fig. 3). These gratings are Bragg reflectors, and wavelength of incident light that would be reflected from a particular point along the length of grating depends on the grating pitch at that point.

As is evident from the CFBG of Fig. 2, shorter wavelengths (or higher frequencies) are reflected off nearer points and longer wavelengths (lower frequencies) are reflected off farther points. A sensor of this kind can be very effectively used to measure position, velocity and pressure resulting from a blast. It can also be used to measure temperature due to temperature dependence of wavelength. It measures total light return of an incoherent or a broadband light source reflected off a linearly-CFBG sensor in contact with the explosive (Fig. 3).

The circulator isolates input and output optical signals. Assuming flat emission and detection profiles over CFBG length, the operation depends

on the linear relationship between wavelength-integrated reflectivity and CFBG length. Complete instrumentation comprises a CFBG sensor and an interrogation system.

Spectroscopic sensors

Both atomic emission and ultraviolet (UV) absorption spectroscopy have been used for the measurement of fireball temperature of explosive blasts. While emission spectrometry is effective for the measurement of fireball surface temperature, UV absorption spectroscopy is effective for the measurement of temperature of fireball core. Atomic emission spectrometry for fireball temperature measurement makes use of intensity of light emitted at a particular wavelength to determine temperature (Fig. 4).

In the case of optically-dense fireballs, temperatures derived from emission measurements may be representative of only the outer edge of the fireballs. Preferred method in such cases is UV absorption spectroscopy, which provides quantifiable information on the temperature of the core. Dye laser pumped by third harmonic of Nd-YAG laser is generally used for UV absorption spectroscopy.

Proximity sensors

An optronic proximity sensor is a type of laser rangefinder designed to accurately measure relatively smaller distances to target, as compared to those encountered in conventional laser rangefinders used in tactical battlefield applications, such as observation, surveillance and fire control.

A conventional battlefield laser rangefinder measures distance to target. Range information is displayed to the observer when the device is used in observation and surveillance role, or fed to a fire control computer in a fire-control application. In the case of a proximity sensor, processing circuitry generates a command signal when distance to target is equal to the preset distance value within a certain specified tolerance.

Command signal, in turn, could be used to perform a variety of control

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